

The Duality of Risk and Safety in the Bitcoin Market

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Abstract

This paper analyzes the hedging properties of Bitcoin and discusses the specific cases in which it may achieve this goal. I hypothesize Bitcoin will react in macroeconomic crises surrounding money and banking as a weak safe-haven asset, working as a hedge against systemic risk in the economy, while reacting as a risk-on asset in all other crises. I analyze this hypothesis with two comparative case studies: the Covid-19 pandemic and the collapse of Silicon Valley Bank in 2023. My results suggest Bitcoin does indeed serve as a weak safe haven, or at least a successful diversification, in crises surrounding money and banking.

1 Introduction

Why is it that Bitcoin was built to mimic gold’s properties as a store of value and yet experiences extreme volatility and 10 percent price changes in mere hours? Over the last decade this has been the subject of much discussion. Gold is mined in the ground—an expensive practice which, given current gold stock in circulation, keeps the market from being flooded with new supply. Similarly, Bitcoin is “mined” using vast amounts of computing power to ensure its own market will not be flooded with new supply (Nakamoto, 2008). Clearly, Bitcoin was inspired in part by the design of safe-haven assets, such as gold, and certainly shares some similarities – such as its inherent scarcity and high stock-to-flow ratio (among others) – which are discussed at length in the seminal book *The Bitcoin standard: The decentralized alternative to central banking* (Ammous, 2018). And yet, its volatility suggest there are other mechanisms at play (Klein et al. 2018, Al-Khazali et al. 2018). So then, is Bitcoin a safe-haven asset, risk-on asset, or something else entirely?

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To begin with I will define terms. In this paper I define a risk-on asset as an asset which falls in price in times of market uncertainty. Following the definitions of Baur & Lucey (2010), I define a safe-haven asset as "an asset that is uncorrelated or negatively correlated with another asset or portfolio in times of market stress or turmoil." Baur & Lucey, then, have two separate categories for safe havens which I call "weak" and "strong" safe-haven assets. The former are assets which are simply "uncorrelated" with another asset in times of market distress – the latter are assets which are specifically "negatively correlated" with another asset during market distress. Notice these definitions are relative to the asset of comparison, a safe haven is uncorrelated or negatively correlated with "another asset." In this paper, I define safe haven in relation to risk-on assets. Finally, it is also important to note Baur & Lucey (2010) are vague in their definition of "uncorrelated" – this, as the reader will soon discover, is important to the analysis of this paper.

Bitcoin emerged as a novel asset out of the rubble of the 2008 financial crisis. Since then, researchers have attempted to classify Bitcoin into a traditional asset category. This has proved to be a difficult task given the unique, and often counterintuitive, reactions Bitcoin exhibits in response to market events. However, this has not deterred researchers from finding similarities between Bitcoin and other assets and making claims as to its nature. Specifically, a fierce debate has emerged between two camps in the literature: is Bitcoin a risk-on asset or a safe-haven asset (Kang et al, 2025)?

The literature in support of Bitcoin as a speculative asset emerged early in Bitcoin's history. Klein et al. (2018) find that over the time horizon from 2011 to 2017 Bitcoin and gold were not closely correlated in volatility or times of market distress. Over a similar time horizon Al-Khazali et al. (2018) find Bitcoin reacted opposite to gold in the face of macroeconomic surprises. As gold is considered the standard for safe-haven assets, they conclude Bitcoin is not a safe-haven. In Bitcoin's more recent history, Muzic & Gržeta (2022) study seven macroeconomic shocks (Non-Farm Payrolls, Retail Sales, Inflation Rate, Core Inflation Rate, Durable Goods Orders, Personal Spending, and Personal Income) and found that Bitcoin reacts more strongly to these shocks than the S&P 500, implying it is not a safe-haven but rather a risk-on asset.

On the other side of the literature, researchers have found significant similarities between Bitcoin and safe-haven assets. Dyhrberg (2016) found Bitcoin has some hedging capabilities similar to gold and the dollar. Specifically, the paper suggests Bitcoin belongs somewhere between gold and the dollar in its store of value advantages. Similarly, Corbet et al. (2020) find that for certain macroeconomic reports (unemployment and durable goods), Bitcoin responds as a safe-haven asset.

The literature is lacking in one clear way. As stated above, Bitcoin is an entirely new asset that does not perfectly fit into one asset class or another. The problem in the existing literature is that each paper attempts to show why Bitcoin does indeed either fit into the classification of a risk-on asset or a safe-haven asset. This paper suggests it is incorrect to strictly categorize Bitcoin in one camp or the other and seeks to present an alternative approach which looks at Bitcoin within specific types of macroeconomic shocks to see if there is a pattern in how Bitcoin reacts. This paper will do this by looking at the comparative cases of the banking collapses of March 2023 and the Covid-19 Crisis of March 2020. Using a Dynamic Conditional Correlation Generalized Autoregressive Conditional Heteroskedasticity (DCC-GARCH) model this paper will show that Bitcoin does not react in a uniform way to all macroeconomic shocks but rather can react either as a risk-on asset or safe-haven asset depending on the type of shock.

Utilizing the DCC-GARCH model, I estimate the variation of Bitcoin and the S&P 500 individually and the correlation between them over the period from 01/01/2020 to 11/07/2025. Using this data I estimate if there is a change in the correlation between these two assets in the wake of two separate crises: Covid-19 and the collapse of Silicon Valley Bank. I find there is a small positive change in the correlation of the two assets after the Covid-19 pandemic began, but I find a negative and strongly significant change on the correlation between the two assets after the collapse of Silicon Valley Bank. This suggests that while Bitcoin did react as a risk-on asset in the wake of the Covid-19 pandemic, it did not react as a tradition risk-on asset in the banking crisis of 2023.

2 Theory

It is well-documented in the academic literature that in times of macroeconomic shocks, uncertainty in a society increases (Baker et al., 2016). When uncertainty changes, individuals will tend to reallocate their resources based on their preferred risk levels. Therefore, when macroeconomic shocks take place, increasing perceived risk, individuals will move their assets directionally towards less risky assets. This is referred to as a flight to quality (Caballero & Krishnamurthy, 2008). The question of what constitutes as a "quality" asset is clearly an important concept to tackle in this paper. I argue that in most macroeconomic shocks, Bitcoin is not viewed as a quality asset – meaning there will be no flight to Bitcoin in these shocks. However, I argue in this paper that when the macroeconomic shock pertains to money and banking, Bitcoin can be perceived as a quality or safe-haven asset. The hypotheses below more clearly state the ideas in this paper:

H_1 : In the face of most macroeconomic shocks, Bitcoin will not be viewed as a safe-haven asset, but rather a risk-on asset.

H_2 : In the face of a money or banking macroeconomic shock, Bitcoin will be viewed as a safe-haven asset, not a risk-on asset.

This section will explain to the reader the theoretical reasoning for why these hypotheses are reasonable.

I will begin by discussing the reasons H_1 may be a reasonable hypothesis. Like all new technological innovations, there are risks associated with Bitcoin. The financial literature generally classifies financial assets as having quantifiable risks (Sharpe, 1964; Black & Scholes, 1973). Bitcoin – being still in its infancy – contains great risks, and all investors are taking that risk upon themselves. If this is true, it is reasonable to assume that Bitcoin would react in the marketplace as a risk-on asset. This means that in times of macroeconomic crises – when uncertainty grows – we would expect Bitcoin to follow closely with other risk-on assets, such as the S&P 500. If this

framework holds, it appears H_1 will hold as well.

H_2 is important, however, because in certain macroeconomic shocks Bitcoin may react contrary to risk-on assets. This is because Bitcoin serves as a Schelling point – a type of rallying cry – in macroeconomic crises surrounding money and banking. To understand this idea we must get to the core of the Bitcoin experiment.¹ To do this, we must go back to the genesis block of Bitcoin – the first transaction ever settled on the Bitcoin blockchain – which had a message attached: "The Times 03/Jan/2009 Chancellor on brink of second bailout for banks." Officially, the message was just stating the front page news of that morning as reported in the British newspaper The Times. The deeper message that Satoshi Nakamoto (the creator of Bitcoin) left for future generations, however, was just as easy to see: *traditional banking is failing – here is the alternative*.

This reiteration of Bitcoin's origins hopefully gives enough of the context needed to understand the zeitgeist of the early Bitcoin community. While Bitcoin has transcended this simple purpose as its popularity has ballooned, this central theme of Bitcoin as an alternative to tradition banking and finance has stayed top of the mind. In other words, I argue in this paper Bitcoin reacts as a Schelling Point in crises surrounding money and banking.

A Schelling Point – or focal point – is a concept developed by Thomas Schelling (Schelling, 1960). The idea is this: in a coordination game where there is limited communication, a stable equilibrium solution can emerge if the individuals in the game all recognize a conspicuous solution that allows for coordination without communication. The famous example is two individuals who agree to meet the next day in New York City. If they forget to set a meeting place, one might suspect a breakdown in equilibrium, as New York City is so large there is a near zero chance they will find each other the next day. Schelling argues, however, that certain locations, such as under the clock at Grand Central Station, stand out as such an obvious meeting place that both individuals understand – without communication – that they should meet there. In Schelling's own words, the Schelling (focal) point sets "each person's expectation of what the other expects him to expect to be expected to do" (Schelling, p. 57, 1960).

¹A brief history of Bitcoin is given in the Appendix to provide more context if needed.

In times of market distress, risk and uncertainty grows, leading to a need for society to re-establish equilibrium for its risk tolerance in financial markets. However, individuals in financial markets have only a limited ability to communicate in the fast-moving crisis. Therefore, when the systemic risk of the banking industry increases and there is limited communication, Schelling's game plays out with individuals attempting to coordinate without communication. For a large segment of the population (generally younger individuals), in times of crises surrounding money and banking, I argue Bitcoin stands out as a Schelling Point – an asset that these individuals understand others will acquire as a hedge against the risk of tradition money and banking. This then makes it the obvious choice of quality for them as well. Bitcoin then, because of its cultural significance in society, leads to mutual expectation in these financial crises.

Therefore, in other macroeconomic crises outside of money and banking, Bitcoin will be expected to react as a traditional risk-on asset. In crises surrounding money and banking, however, it may be that Bitcoin reacts as a type of "systemic-risk" safe-haven asset due to its cultural acceptance as the default mechanism to escape tradition markets. In other words, Bitcoin is inversely related to the stability of traditional money and banking markets.

To make this more clear, let us go through a comparative example of Bitcoin in two crises. The first crisis, for illustrative purposes, is a poor jobs report. In this case, an economic contraction is expected, leading stocks lower as people adjust expectations for future uncertainty. We would also expect Bitcoin's price to fall as it operates similar to other risk-on assets in this crisis. Now let us consider a separate crisis, such as the U.S. government defaulting on its debt. In this case, what is considered a "quality asset" may change drastically (for example, Treasuries, which typically fall under the safe-haven category will be flooded with risk). Similarly, we may expect demand for Bitcoin to rise as it serves as a Schelling Point in this time of "systemic risk," representing an alternative to traditional money and banking. While this example is dramatic, it illustrates the core of the hypothesis.

In this paper, I test my hypotheses with two crises. I test H_1 with the Covid-19 crisis. In this case, there is limited to no systemic risk, only increased uncertainty in the economy. I test H_2 with

the Banking Crisis of 2023. In this case, it is reasonable to hypothesize an increase in systemic risk as individuals become aware of the weaknesses in the banking sector.

3 Empirical Analysis

3.1 Data

To perform my analysis, I collected daily return data for Bitcoin and the S&P500 from Yahoo! Finance (n.d.) from 01/01/2020 to 11/07/2025. To perform this collection, I utilized the `yfinance` package in Python (Aroussi, 2025). Summary statistics of the data are given in Table 1. Figure 1 shows the price performance for the two assets over the period of analysis to give an idea of their trends.

Table 1: Summary Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
BTC Price	1,471	46,153	31,044	4,971	124,752
S&P Price	1,471	4,531	978.93	2,237	6,891
Return_BTC	1,471	0.1832	3.9522	-46.4730	19.1527
Return_S&P	1,471	0.0493	1.3330	-12.7652	9.0895
Variance_BTC	1,471	17.0066	18.1968	5.8688	317.80
Covariance	1,471	1.8407	3.0527	0.0287	45.9680
Variance_S&P	1,471	1.7006	3.5038	0.3761	48.1549
DCC	1,471	0.3543	0.1019	0.0081	0.5969

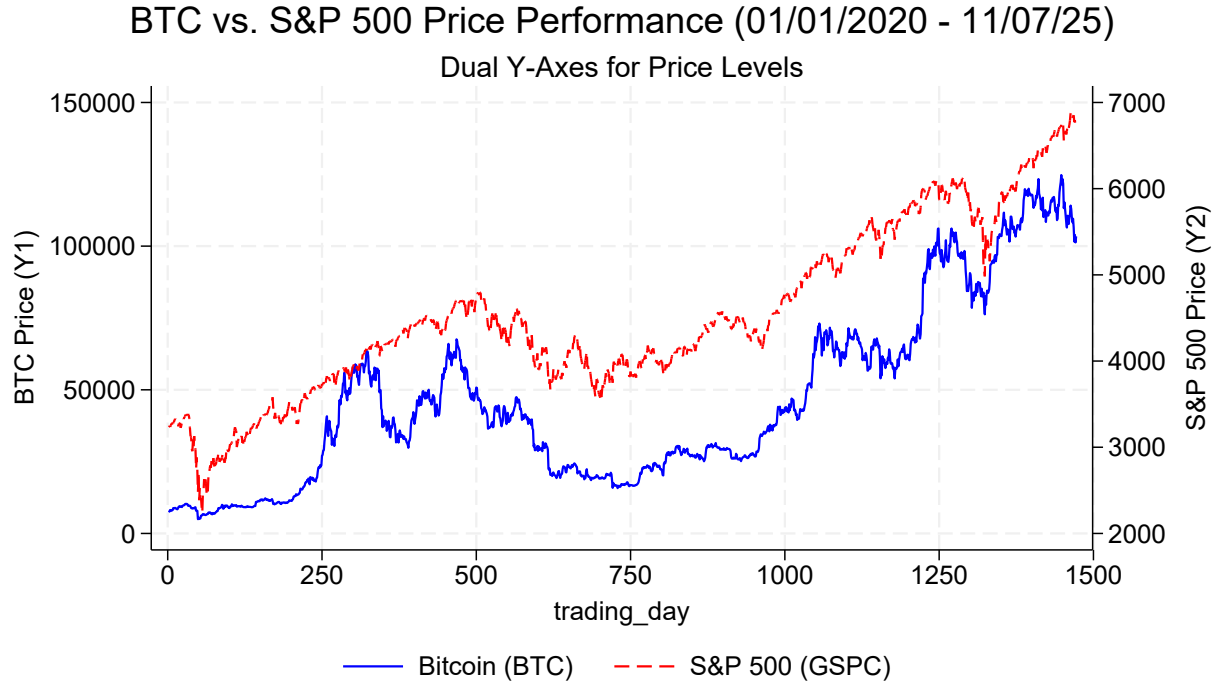


Figure 1

3.2 Empirical Model

For my analysis I utilize a DCC-GARCH model (Engle, 2002). A Dynamic Conditional Correlation Generalized Autoregressive Conditional Heteroskedasticity (DCC-GARCH) model is used to estimate the variation of two individual assets (two separate GARCH models) and then runs a model on top of these two GARCH models to find the correlation between the two assets (the DCC part). In doing so, this model gives me the individual variation of the assets, as well as the correlation between them. This model is perfect for my research as it provides a daily correlation estimate over the entire data horizon, which allows me to analyze how this correlation changes (or remains constant) over time.

I utilize this model to answer a simple question: do Bitcoin and the S&P 500 remain just as correlated in the aftermath of a macroeconomic shock or does the relationship change? I analyze this question for two discrete events. The first is the Covid-19 pandemic, which I input to my data as a crisis dummy variable beginning on March 16, 2020 (the day the United States began the "15

days to slow the spread” lockdown). The second is the banking crisis of 2023, which I input in my data as a second crisis dummy variable beginning on March 10, 2023 (the day Silicon Valley Bank collapsed).

3.3 Results

I study the effects over several post-treatment windows: 30, 50, 100, and 200 days. Figure 2 and Figure 3 show the dynamic conditional correlation between the two assets in a 50 day pre- and post-treatment window. The vertical black-dotted lines indicate the day of the crisis. As Figure 2 shows, in the Covid-19 pandemic, Bitcoin and the S&P 500 remain closely correlated in the 50 days after the crisis begins – if anything the relationship grows from the pre-treatment period (maybe implying some anticipation of the crisis). Figure 3 tells a different story, however. In the case of the banking crisis of 2023 there is a steep drop in the correlation of Bitcoin and the S&P 500.

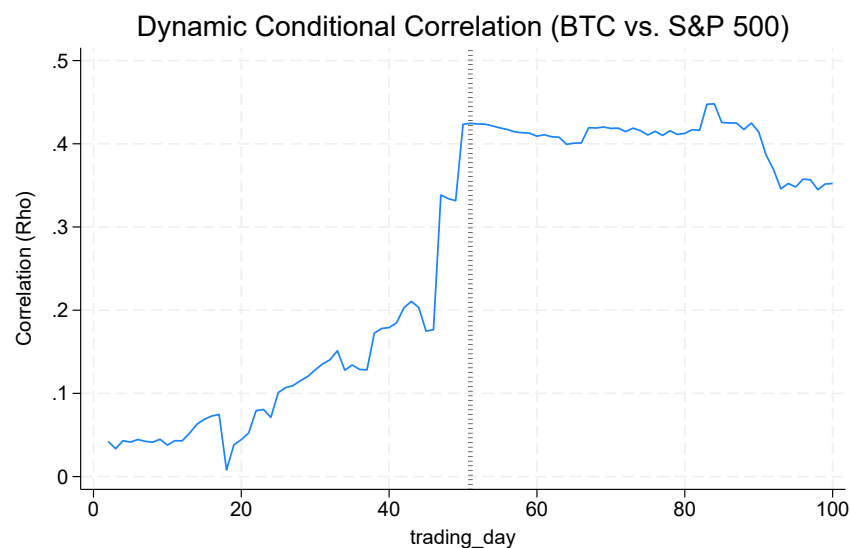


Figure 2: 2020 Covid-19 Crisis

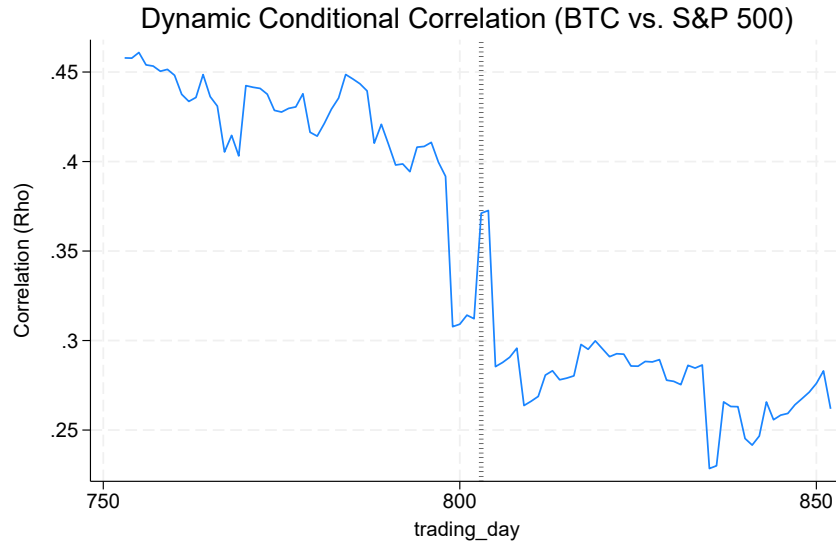


Figure 3: 2023 Banking Crisis

While the graphs seems to suggest two separate relationships in the crises, a more robust result must be tested. To test my hypothesis I regress the two crisis dummy variables I described above on the dynamic conditional correlation output. It should be noted, this regression runs over the entire period of data collected, not only the period shown in the graph. The output is shown in Table 2 and the hypothesis testing in Table 3. The output tells us several things. First, the constant term tells us that over the period from 01/01/2020 to 11/07/2025 the average correlation between these assets was around 0.35. The Crisis 1 variable tells us that in the aftermath of Covid-19, correlation increased – meaning Bitcoin became more correlated to risk-on assets (proxied by the S&P 500). The Crisis 2 variable tells us that in the days after the collapse of Silicon Valley Bank, the correlation fell by between 0.06 to nearly 0.10 (depending on specification of crisis length), meaning Bitcoin became less correlated with risk-on assets (proxied by the S&P 500).

Table 2: Robustness Check: Conditional Correlation ($DCC_{BTC, GSPC}$) during Crisis Windows

Window Length	30-Day	50-Day	100-Day	200-Day
Panel A: Baseline and Goodness-of-Fit				
Intercept ($\hat{\beta}_0$): Normal DCC	0.3543***	0.3552***	0.3592***	0.3670***
Observations	1,471	1,471	1,471	1,471
R^2	0.0150	0.0272	0.0613	0.0942
Panel B: DCC Shift (ΔDCC) from Baseline				
Crisis 1 Shift (β_1): Banking Shock (Day 803)	−0.0636***	−0.0764***	−0.0957***	−0.0915***
<i>t</i> -statistic	(−7.09)	(−8.34)	(−9.86)	(−7.53)
Crisis 2 Shift (β_2): Typical Crisis (Day 51)	0.0599***	0.0499***	0.0231*	−0.0023
<i>t</i> -statistic	(8.67)	(4.79)	(2.07)	(−0.17)
Panel C: Resulting DCC (Correlation during Crisis)				
Banking Crisis DCC ($\hat{\beta}_0 + \beta_1$)	0.2907	0.2788	0.2635	0.2755
Covid-19 Crisis DCC ($\hat{\beta}_0 + \beta_2$)	0.4142	0.4051	0.3823	0.3647

Note: *t*-statistics are in parentheses and calculated using Newey-West standard errors (lag=5).

Significance levels: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 3: Hypothesis Testing

Window Length	30-Day	50-Day	100-Day	200-Day
Test 1: Equality of Crisis Impacts Hypothesis				
$H_0 : \beta_0 + \beta_1 = \beta_0 + \beta_2$				
<i>p</i> -value	0.000	0.000	0.000	0.000
Test 2: Uncorrelated Hypothesis				
$H_0 : \beta_0 + \beta_1 = 0$				
<i>p</i> -value	0.000	0.000	0.000	0.000

These results paint an interesting picture. As Panel C of Table 2 shows, in the Covid-19 pandemic, the correlation between Bitcoin and risk-on assets sits around 0.40, while in the banking crisis of 2023 this correlation falls to around 0.27. This is a large drop in correlation – a change representing two assets going from moderately correlated to weakly correlated. Test 1 in Table 3 tests if the reaction to the two shocks are statistically indistinguishable from each other. The results confirm Bitcoin is not reacting uniformly to macroeconomic shocks, but rather can react in opposing directions depending on the type of shock. These results also shed light on our interpretation of "uncorrelated" as defined by Baur & Lucey (2010). As Test 2 of Table 3 shows, Bitcoin and the S&P 500 cannot be characterized as entirely uncorrelated during the banking crisis (the correlation coefficient is statistically different from zero during the crisis), suggesting a correla-

tion coefficient of 0.27 is not weak enough to consider the assets uncorrelated. Therefore, to call Bitcoin a safe-haven may be too strong given the evidence if we adhere to a strict definition of "statistically uncorrelated." In this case, Bitcoin is not a true safe-haven asset but rather something akin to a "crisis diversifier." Baur & Lucey (2010) define a diversifier asset as, "an asset that is positively (but not perfectly correlated) with another asset or portfolio on average." Bitcoin clearly appears at the very least as a diversifier asset in the face of a money and banking crisis. These results are consistent with our hypothesis that when the macroeconomic shock is related to money and banking, Bitcoin may react – in the short term – as a weak safe-haven asset, while still trading the remainder of the time (and in the majority of macroeconomic shocks) as a risk-on asset.

4 Conclusion

The debate in the academic literature (Kang et al. 2025, Muzic & Gržeta 2022, Dyhrberg 2016) between those who view Bitcoin as a risk-on asset and those who view Bitcoin as a safe-haven asset is far from being settled. In this paper I suggest, however, that both camps are going about the debate erroneously and that Bitcoin as an asset deserves to be studied with more nuance. Specifically I hypothesize Bitcoin is neither a risk-on nor safe-haven asset concretely, but that it may react in either direction depending on the crisis at hand. In this paper I analyze the effects of the Covid-19 pandemic and the banking crisis of March 2023 on the correlation of Bitcoin and risk-on assets (proxied by the S&P 500). I find evidence, robust to different crisis window specifications, that Bitcoin does indeed provide at the very least hedging capabilities as a diversifier asset – and may even reach the status of a weak safe haven – in times of crisis in the money and banking markets.

Appendix

The idea of Bitcoin came out of the 2008 financial crisis. It was created in response to what some viewed as an example of crony capitalism, where the central banks of nations were willing to bail out Wall Street but not Main Street. The anonymous creator of Bitcoin referred to himself

as Satoshi Nakamoto. Before the technology itself was created, Satoshi produced a white paper (a document explaining a problem and proposing a solution) (Nakamoto, 2008). The white paper was aimed at solving one problem: the necessity for trusted third parties in financial transactions. For Nakamoto, the problem was, in essence, society's dependence on banks as an intermediary in financial transactions. The solution to this problem, in Nakamoto's own words was "an electronic payment system based on cryptographic proof instead of trust, allowing any two willing parties to transact directly with each other without the need for a trusted third party" (Nakamoto, 2008). The white paper then set down the framework for such a system. While the intricacies of the system are beyond the scope of this paper, the main points were as follows. Transactions would be done peer-to-peer without a third party intermediary. The transactions would be posted on a digital ledger – referred to as a blockchain. To ensure a valid transaction and avoid the need for trust, the blockchain would be continually patrolled and validated by other individuals on the network. The validators are incentivized to do this through a process called "proof of work" which allows them to collect Bitcoin rewards if they correctly perform their duties. Through this process, so long as 50% +1 of the validators on the network are not corrupting the system, it will allow for direct peer-to-peer transaction without any need for trust. Shortly after the white paper was anonymously published by Satoshi Nakamoto, the technology of Bitcoin was successfully created. The first transaction ever settled on the Bitcoin blockchain had a message attached: "The Times 03/Jan/2009 Chancellor on brink of second bailout for banks." Officially, the message was just stating the front page news of that morning as reported in the British newspaper The Times. The deeper message that Satoshi Nakamoto left for future generations, however, was just as easy to see: *traditional banking is failing – here is the alternative.*

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